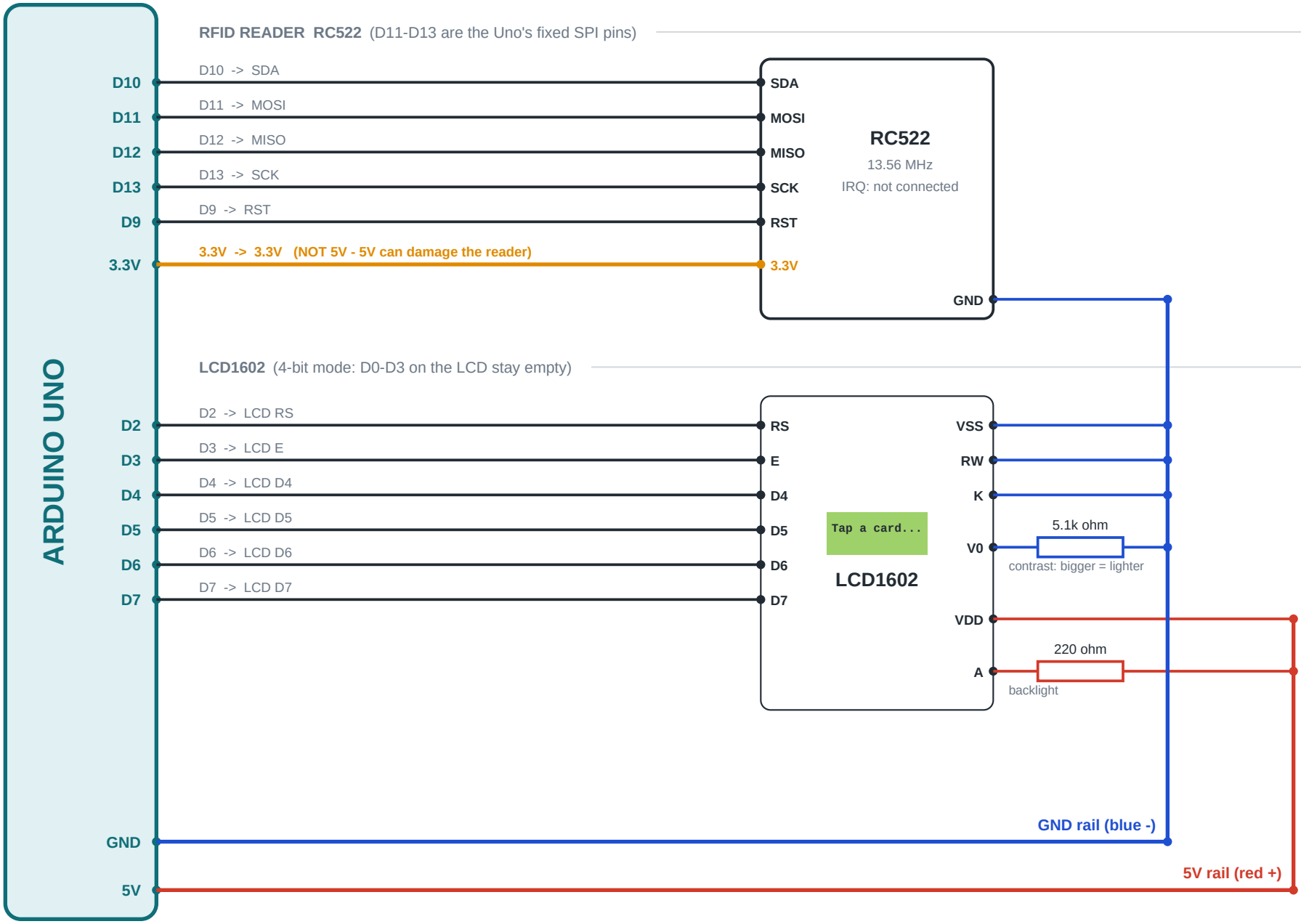
RFID Music Box - Phase 1 Wiring

Arduino Uno | RC522 RFID reader + LCD1602 | Sketch: src/main.cpp

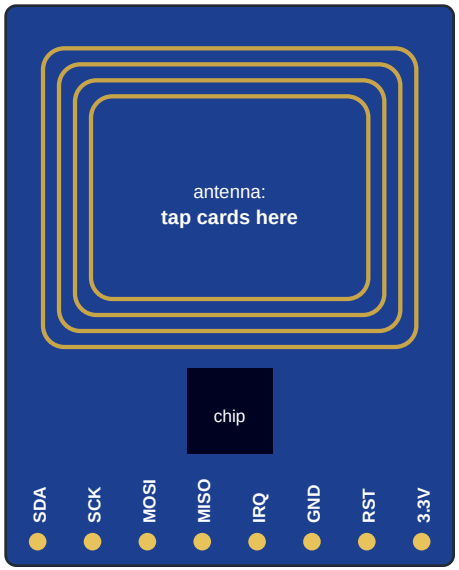


Lines that cross without a dot are not connected. The RC522's 3.3V is the only wire that does not use a rail.

RFID Music Box - Phase 1 Pin Guide

How to find each pin on the real boards. Pin names are printed on the boards; colours match page 1.

RC522 RFID READER (component side up)

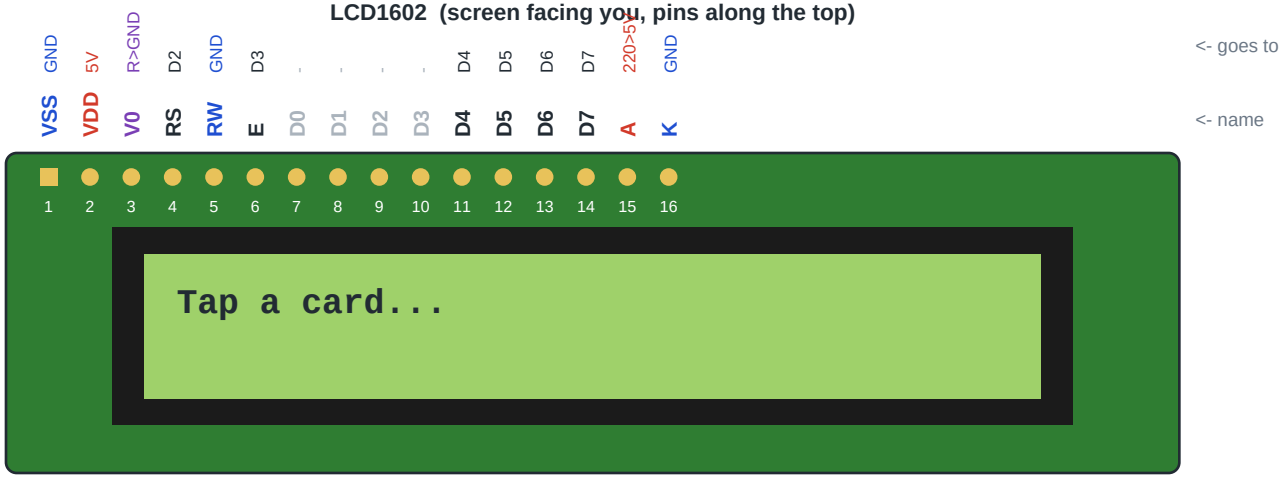


D10 D13 D11 D12 - GND D9 3.3V

goes to (Uno)

RC522 pin	Uno pin	Notes
SDA	D10	Also called SS or NSS
SCK	D13	Hardware SPI
MOSI	D11	Hardware SPI
MISO	D12	Hardware SPI
IRQ	-	Leave empty
GND	GND rail	
RST	D9	
3.3V	**3.3V**	**Never 5V**

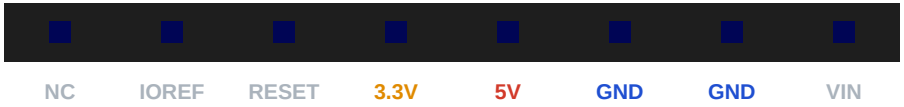
LCD1602 (screen facing you, pins along the top)



Pin 1 (VSS) is on the left and often has a square pad. Some boards print only 1 and 16.

#	LCD pin	Goes to	#	LCD pin	Goes to
1	VSS	GND rail	9	D2	Leave empty
2	VDD	5V rail	10	D3	Leave empty
3	V0	Contrast resistor, then GND rail	11	D4	Uno D4
4	RS	Uno D2	12	D5	Uno D5
5	RW	GND rail	13	D6	Uno D6
6	E	Uno D3	14	D7	Uno D7
7	D0	Leave empty	15	A	220 ohm, then 5V rail (backlight)
8	D1	Leave empty	16	K	GND rail (backlight)

UNO POWER HEADER (8 pins next to A0-A5, on the opposite edge from D0-D13)



3.3V feeds the RC522 (the Uno makes it itself, no resistor needed). **5V** and one **GND** go to the breadboard's red (+) and blue (-) rails.

RFID Music Box - Phase 1 Parts & Connections

Everything on page 1 as lists, plus a few tips.

Parts

Qty	Part	Notes
1	Arduino Uno + USB cable	Powers everything.
1	RC522 RFID module, card, key fob	From the kit. Solder the header pins if they are loose.
1	LCD1602 module	16 pins along the top edge.
1	5.1k ohm resistor	Green, brown, red (the kit's 5K1). LCD contrast.
1	220 ohm resistor	Red, red, brown. LCD backlight.
1	Breadboard	Uno 5V to the red (+) rail, GND to the blue (-) rail.
~22	Jumper wires	Female-to-male for the RC522 if it has header pins.

Connections

Uno pin	Goes to	Notes
D10 / D9	RC522 SDA / RST	
D11 / D12 / D13	RC522 MOSI / MISO / SCK	Hardware SPI.
3.3V	RC522 3.3V	Not 5V. Power header, between RESET and 5V.
D2 / D3	LCD RS / E	
D4 - D7	LCD D4 - D7	Same number on both ends.
GND rail	RC522 GND, LCD VSS, RW, K	RW to GND = write-only.
GND rail	5.1k ohm, then LCD V0	Contrast.
5V rail	LCD VDD; 220 ohm, then LCD A	Power and backlight.

If something is wrong

Text hard to read: it's the contrast resistor on V0 (5.1k worked here). Text faint or missing: use a smaller one (2k, 1k). Grey or solid blocks behind the text: use a bigger one (10k).

Solid blocks that never change: the sketch isn't talking to the LCD. Check RS, E and D4-D7, and that the upload worked.

Nothing when you tap a card: check the reader gets 3.3V, the SPI wires are on D10-D13, and the header is soldered. Hold the card flat, within 2 cm.

LCD says RFID not found: the Uno asked the reader for its version and got no answer. Check all 7 RC522 wires, especially 3.3V and GND, then press the Uno's reset button.

Serial Monitor shows gibberish: set it to 9600 baud (pio device monitor already does).

Card shows once, then not again: by design. Lift it off the reader and tap again.

ID changes on every tap: phones and some bank cards use random IDs. Use the kit's card, fob or MIFARE/NTAG tags.

What it does

While idle the LCD shows **Tap a card...** On a tap the LCD shows **Card:** and the ID for 3 seconds, and the Serial Monitor prints **Card UID: A3 4F 12 9C**. Write each ID down: phase 2 maps them to music folders. In phase 2, cards in cards.h play music and unknown cards show **New card:** and their ID instead.